



Computer Networks

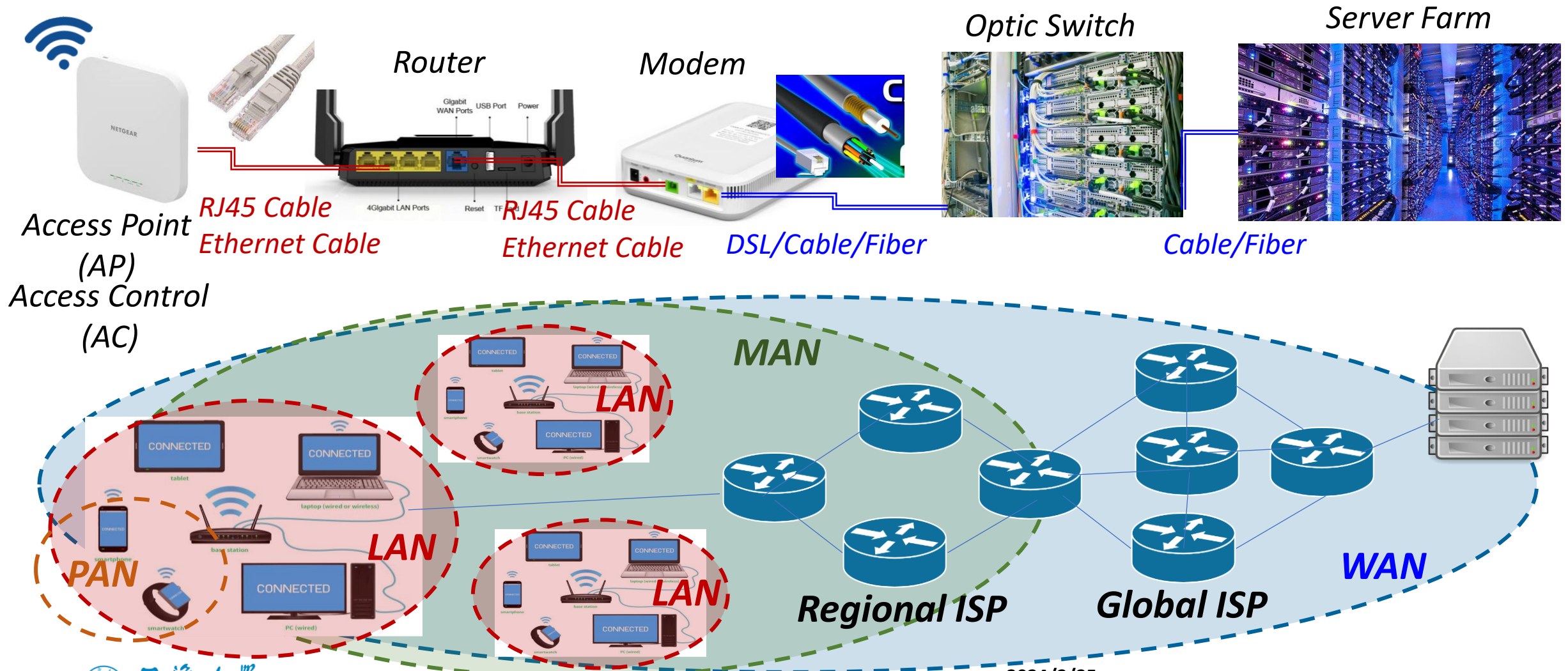
计算机网络 (42034401)

2023-2024 学年第二学期 (24-Spring)

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Connecting to the Internet





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Syllabus

A Brief Introduction (1/2)

- **A Required Course in Software Engineering, 3 CH, 51 lectures**

- **Fri.(Every Week) 13:30-16:15**

- **Objectives/ What to learn :**



A bit of **history**: Why should we learn this course? What is already there?



Some **concepts**: What is a computer network? **Layered models, protocols, interfaces, services, and applications...**



The '**beef**': **How does every part work?** Functions and frequently-used protocols in each layer.



A bit of '**engineering**': How to apply the algorithms/mechanisms?

A Brief Introduction (2/2)

- **Textbook**

- Andrew S. Tanenbaum and David J. Wetherall, *Computer Networks (5th Edition)*, Pearson Education Inc. 2011

- 机械工业出版社 **Bottom-up** 'Protocol Approach'

- **Other References**

- James F. Kurose and Keith W. Ross, *Computer Networking, A Top-down Approach (7th Edition)*, Pearson Education Inc. 2017
- Larry L. Peterson and Bruce S. Davie, *Computer Networks, A Systems Approach (6th Edition)*, Elsevier Inc. 2020
- 谢希仁, 《计算机网络》, 电子工业出版社, 2021

Agenda (Tentative)

Chap.	Content	Lecture	Exercise	Subtotal
1	Introduction	5	1	6
2	Physical Layer	5	1	6
3	Datalink Sublayer	4	1	5
4	MAC Sublayer	4	1	5
5	Network Layer	6.5	1.5	8
6	Transport Layer	9	2	11
7	Application Layer	5	1	6
8	Security + Review	3	1	4
Total		41.5	9.5	51

Score and Grades

- **Attendance: 考勤 10%**
 - **Homework: 作业 30%**
 - Discussion: 课堂提问、班级群讨论和回答同学问题 6%
 - Individual Homework: 个人作业提交(*3) 24% (作业迟交记一半分数)
 - **Final Exam: 期末考试 60%**
 - Closed-book, closed-notes 闭卷
 - **Academic Integrity: 抄袭等学术不端行为直接记0分**
 - **Contact: 有问题请联系**
 - 王洁 wangjie_tongji@tongji.edu.cn 济事楼 507
 - Walk-in Office Hour: Fri. 16:30-17:00, 其他时间请预约
 - 通过canvas (推荐使用)、课程群 (紧急消息) 或邮件 (个人相关) 联系
- 邮件标题中注明【CN-24S】并写清学号、姓名等信息

You are expected to ... after this course

- *Know why Internet looks like its current status, and how it works*
- *Understand mechanisms in the TCP/IP stack*
- *Write a client-server socket program*



Or more practically,

100 Pass the final exam! *easy*



Pass an interview when asked about a 3-way handshake, slow-start, throughput, CSMA/CD, CRC, NAT, IP tunneling, etc. emm...OK



Configure and troubleshoot your router at home.

Trust me, it is hard!!!

But don't you worry.

- **My English is not good. Will I have difficulty learning this course?**
 - The *textbook* is easy to understand. (Sometimes you will even find it verbose)
 - High school English is enough to understand the *lecture*, except...
 - Major challenge: **new concepts** -> figures + explanation
 - *Q&A* in class and on Canvas
- **Will the exam be hard?**
 - The exam will be *exactly as difficult* as the other classes.
 - All problems in the exam will have been *covered in this class*.

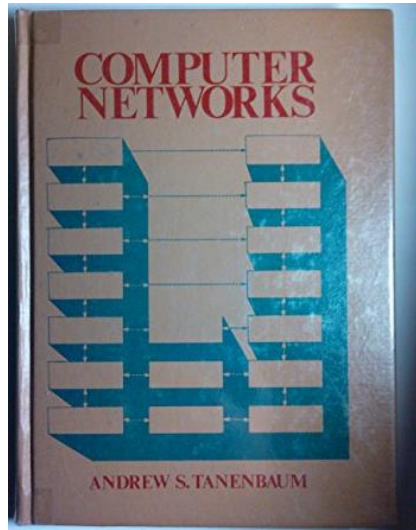


CONCERN



Some Tips on Learning

- **Complex, but not difficult** -> Build your own **architecture**.
- **Key: the protocol hierarchy – the software part in networking**



Computer Networks (1st Ed.) by A. Tanenbaum

You will see this figure throughout this semester.

- **Understand ‘WHY’, and naturally you will know ‘HOW’.**

A Favor I need to ask from you

- I tend to speak fast when I am nervous/excited.
- I got excited easily when I give a lecture. 🌟🌟
- So, please **raise your hand** when I (unintentionally) speak too fast or you have a question, and I will slow down.

